

Bachelor of Science in Computer Science

Birla Institute of Technology & Science, Pilani

CURRICULUM

Foundation courses

COURSE NO.	COURSE TITLE	UNITS
MATHEMATICS FOUNDATION		
BCS ZC219	Discrete Mathematics	3
BCS ZC230	Linear Algebra and Optimization	3
BCS ZC233	Probability and Statistics	3
BCS ZC112	Introduction to Logic	2
SCIENCE & ENGINEERING FOUNDATION		
BCS ZC223	General Biology [§]	3
BSC ZC240	General Physics [§]	3
BCS ZC111	Basic Electronics	2
HUMANITIES FOUNDATION		
BCS ZC151	Writing Practice	3
BCS ZC113	Online Social Media [#]	2
BCS ZC114	Video Games - Technology and Social Impacts [#]	2
OTHER COURSES		
BCS ZC220	Environmental Studies	3
BCS ZC229	Introduction to Economics [@]	3
BCS ZC235	Science, Technology and Modernity [@]	3

- [§]One course must be chosen from this list of courses (Foundation Option 1 in the program chart)
- [#]One course must be chosen from this list of courses (Foundation Option 2 in the program chart)
- [@]One course must be chosen from this list of courses (Foundation Option 3 in the program chart)

Core Courses

COURSE NO.	COURSE TITLE	UNITS
BCS ZC313	Introduction to Programming	4
BCS ZC228	Introduction to Computing Systems	3
BCS ZC216	Computer Systems and Performance	3
BCS ZC215	Command Line Interfaces and Scripting	3
BCS ZC311	Data Structures and Algorithms	4
BCS ZC212	Algorithm Design	3
BCS ZC316	Object Oriented Programming	4
BCS ZC317	Relational Databases	4
BCS ZC234	Programming for Mobile Devices	3
BCS ZC238	Web Programming	3
BCS ZC232	Operating Systems	3
BCS ZC236	Software Design Principles	4
BCS ZC214	Building Database Applications	3
BCS ZC211	Software Development Practices	3
BCS ZC231	Network Programming & Client-Server Programming	3
BCS ZC222	Formal Languages and Applications	3

Electives

COURSE NO.	COURSE TITLE	UNITS
BCS ZC224	Graphs and Networks	3
BCS ZC213	Automata and Computability	3
BCS ZC221	Experimental Algorithmics	3
BCS ZC227	Introduction to Bioinformatics	3
BCS ZC217	Data Visualization	3
BCS ZC312	Introduction to Data Analytics	4
BCS ZC315	Multicore and GPGPU Programming	4
BCS ZC237	TCP/IP and Internet	3
BCS ZC226	Information Security	3
BCS ZC225	Human Computer Interaction	3
BCS ZC218	Designing Multimodal Interfaces	3
BCS ZC314	Modern Databases	4

Semester breakup:

FIRST SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
BCS ZC313	Introduction to Programming	4
BCS ZC219	Discrete Mathematics	3
BCS ZC230	Linear Algebra and Optimization	3
BCS ZC228	Introduction to Computing Systems	3
BCS ZC111	Basic Electronics	2
BCS ZC239	Writing Practice	3
Units		18

SECOND SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
BCS ZC311	Data Structures and Algorithms	4
BCS ZC316	Object Oriented Programming	4
BCS ZC215	Command Line Interfaces and Scripting	3
BCS ZC233	Probability and Statistics	3
BCS ZC112	Introduction to Logic	2
-	Foundation Option 1	3
Units		19

THIRD SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
BCS ZC212	Algorithm Design	3
BCS ZC317	Relational Databases	4
BCS ZC238	Web Programming	3
BCS ZC236	Software Design Principles	4
BCS ZC216	Computer Systems and Performance	3
-	Foundation Option 2	2
Units		19

FOURTH SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
BCS ZC232	Operating Systems	3
BCS ZC214	Building Database Applications	3
BCS ZC234	Programming Mobile Devices	3
BCS ZC220	Environmental Studies	3
BCS ZC222	Formal Languages and Applications	3
-	Discipline Elective #1	3
Min Units		18

FIFTH SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
BCS ZC211	Software Development Practices	3
BCS ZC231	Network Programming and Client-Server Programming	3
-	Discipline Elective #2	3/4
-	Discipline Elective #3	3/4
BCS ZC241T	Study Project	5
Units		17-19

SIXTH SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
-	Foundation Option 3	3
-	Discipline Elective #4	3/4
BCS ZC428T	Project	10
Units		16-17

SEVENTH SEMESTER:

COURSE NO.	COURSE TITLE	UNITS
-	Discipline Elective	3/4
-	Discipline Elective	3/4
-	Open Elective	3/4
-	Open Elective	3/4
-	Open Elective/ Discipline Elective	3/4
-	Discipline Elective\$	3/4
Min Units		17-20

EIGHTH SEMESTER

COURSE NO.	COURSE TITLE	UNITS
-	Discipline Elective	3/4
-	Discipline Elective	3/4
-	Discipline Elective/ Mini Project	3/5
-	Open Elective/ Discipline Elective	3/4
-	Open Elective/ Discipline Elective	3/4
-	Discipline Elective\$	3/4
Min Units		17-20

- Minimum Units for Semester VII and VIII is 37
- Courses for Discipline Electives can be chosen from the list of Discipline Electives
 - \$ - These Discipline Electives may be dropped if the minimum required units for the respective semesters are met by the choice of the remaining five courses.
 - A specialization is awarded when a student chooses 4 Discipline Elective courses, in their VIIth and VIIIth semesters, for a minimum of 14 units, from a pool of courses in that specialization and a Mini Project for 5 units.
- Open Electives must be chosen from the Open Electives pool.
- A minimum of 4 Discipline Electives (or 4 Discipline Electives and a Mini Project) and 3 Open Electives must be chosen in the VIIth and VIIIth semesters.

The BSCH-CS program design offers three specializations to choose from. The specializations are (1) Full-Stack Development (2) Cloud Computing (3) AIML. The specialization is structured to have four mandatory courses in the specialization area and a mini-project in the domain of specialization. The specialization enhances the depth of knowledge and skills in one of the demanding areas and thereby enhances a learner's employable skills. The learner also has the option to choose courses from across the specializations (on offer) and from disciplinary electives to meet his learning needs. The structure of the program allows the specializations to be added/updated as required.

The Specializations are presented below:

(a) Specialization in Full-Stack Development

Title of the Specialization: Full-Stack Development

COURSE NO.	COURSE TITLE	UNITS	Pre-requisite(s)
BHCS ZC413	Backend and API Development	4	
BHCS ZC419	Frontend Development	3	
BHCS ZC415	Cross-platform Applications	3	Frontend Development
BHCS ZC432	Software Deployment	4	Backend and API Development Frontend Development
BHCS ZC427T	Mini Project	5	

Prerequisite Course to the Specialization: This specialization has no prerequisites.

(b) Specialization in Cloud Computing

Title of the specialization: Cloud Computing

COURSE NO.	COURSE TITLE	UNITS	Pre-requisite(s)
BHCS ZC414	Cloud Computing Fundamentals	3	
BHCS ZC422	Introduction to Networking for Cloud	3	BCS ZC237 - TCP/IP and Internet
BHCS ZC420	Introduction to DevOps for Cloud	4	Cloud Computing Fundamentals
BHCS ZC430	Scalable Services in Cloud	4	Cloud Computing Fundamental
BHCS ZC427T	Mini Project	5	

Prerequisite Course to the Specialization: Students must take the following course as an elective before the end of the sixth semester to pursue Cloud Computing Specialization.

- 1 BCS ZC237 - TCP/IP and Internet

(c) Specialization in AIML

Title of the Specialization: AIML

COURSE NO.	COURSE TITLE	UNITS	Pre-requisite(s)
BHCS ZC421	Introduction to Machine Learning	4	BCS ZC312 - Introduction to Data Analytics
BHCS ZC412	Artificial Intelligence	3	
BHCS ZC417	Deep Learning and Applications	4	Introduction to Machine Learning
BHCS ZC434	Topics in Data Mining	4	BCS ZC312 - Introduction to Data Analytics
BHCS ZC427T	Mini Project	5	

Prerequisite Course to the Specialization: Students must take the following course as an elective before the end of the sixth semester to pursue AIML Specialization.

- 1 BCS ZC312 - Introduction to Data Analytics

Discipline Electives (for Semester VII and VIII)

Title of the specialization: Cloud Computing

COURSE NO.	COURSE TITLE	UNITS	Pre-requisite(s)
BHCS ZC433	Topics in Algorithms and Complexity	4	
BHCS ZC324	Compiler Design	4	BCS ZC237 - TCP/IP and Internet
BHCS ZG512	Network Security	4	Cloud Computing Fundamentals
BHCS ZC321	Software Testing and Automation	3	Cloud Computing Fundamental
BHCS ZC418	Distributed Systems	4	
BHCS ZC319	Natural Language Processing	4	
BHCS ZC423	Introduction to Social Media Analytics	4	BCS ZC237 - TCP/IP and Internet
BHCS ZC416	Cryptography	3	Cloud Computing Fundamentals
BHCS ZG511	Agile Software Processes	4	Cloud Computing Fundamental
BHCS ZC429	Open Source Software	3	
BHCS ZC421	Introduction to Machine Learning#	4	

COURSE NO.	COURSE TITLE	UNITS	PREREQUISITE
BHCS ZC412	Artificial Intelligence#	3	
BHCS ZC417	Deep Learning and Applications#	4	Introduction to Machine Learning
BHCS ZC434	Topics in Data Mining#	4	BCS ZC312 - Introduction to Data Analytics
BHCS ZC414	Cloud Computing Fundamentals\$	3	
BHCS ZC422	Introduction to Networking for Cloud\$	3	BCS ZC237 - TCP/IP and the Internet
BHCS ZC420	Introduction to DevOps for Cloud\$	4	Cloud Computing Fundamentals
BHCS ZC430	Scalable Services in Cloud\$	4	Cloud Computing Fundamentals
BHCS ZC413	Backend and API Development@	4	
BHCS ZC419	Frontend Development@	3	
BHCS ZC415	Cross-platform Applications@	3	Frontend Development
BHCS ZC432	Software Deployment@	4	Backend and API Development Frontend Development

- Mandatory Courses for AIML Specialization

\$ - Mandatory Courses for Full-Stack Development Specialization

@ - Mandatory Courses for Cloud Computing Specialization

Project Course (for Semester VII and VIII)

COURSE NO.	COURSE TITLE	UNITS	PREREQUISITE
BHCS ZC427T	Mini Project*	5	

* - Mandatory for all specializations

Open Electives (for Semester VII and VIII)

	COURSE TITLE	UNITS	PREREQUISITE
BHCS ZC327	Introduction to Calculus	3	
BHCS ZC325	Differential Equations and Applications	3	
BHCS ZC320	Numerical Analysis	3	
BHCS ZC241	Microprocessors, Programming and Interfacing	4	
BHCS ZC328	Introduction to IoT	4	
BHCS ZC244	Accounting for Managers	3	
BHCS ZC322	Corporate Finance	3	
BHCS ZC323	Investment Management	3	
BHCS ZC243	Signals and Systems	3	

Course Descriptions

BCS ZC219 Discrete Mathematics (3 units)

Sets & operation on sets; relations, functions, Recursive functions, sequences and summations, Mathematical Induction, Proof Methods (direct, indirect, proof by contradiction), principle of inclusion & exclusion, pigeonhole principle; Permutations and Combinations; Recurrence Relations; basic algebraic structures and their applications. Introduction to graphs, properties and applications

BCS ZC230 Linear Algebra and Optimization (3 units)

Vector and matrix algebra, systems of linear algebraic equations and their solutions; eigenvalues, eigenvectors and diagonalization of matrices; Formulation of linear programming problems, Simplex method, Big-M method, two phase method, Sensitivity analysis, Revised and Dual Simplex Methods.

BCS ZC233 Probability & Statistics (3 units)

Probability spaces; conditional probability and independence; random variables and probability distributions; marginal and conditional distributions; independent random variables; mathematical expectation; mean and variance; binomial, Poisson and normal distributions; sum of independent random variables; law of large numbers; central limit theorem (without proof); sampling distribution and test for mean using normal and student's t-distribution; test of hypothesis; correlation and linear regression.

BCS ZC112 Introduction to Logic (2 units)

Propositional logic – syntax, semantics, natural deduction, satisfiability & validity, predicate or first order logic – syntax, semantics, proof theory satisfiability & validity, completeness & compactness, proof techniques, Undecidability & incompleteness; Godel's incompleteness theorem; overview of applications in program verification, knowledge representation.

BCS ZC151 Writing Practice (3 units)

Introduction to academic writing – Purpose, Type, Features, Originality, integrity and plagiarism; Paragraph writing – Structure, Development of ideas, Linking paragraphs together, Introduction, opening sentence and conclusion; Elements of writing – Argument and discussion, Cause and effect, Comparison, Generalizations, Problems and solution, Process writing, Visual information, Accuracy in writing, Academic vocabulary, Caution, Conjunction/Linkers/Signposting, Punctuation, Passives, Tense forms, Summarizing and paraphrasing; Writing models - E-mails (formal), Short essays (2-paragraph essay, thesis statement), and Reports (Introduction, Methods, Results, Conclusion, Abstract).

BCS ZC220 Environmental Studies (3 units)

Environment, human population, and industrialization; natural resources and the impact of man-made activities on them; structure and function of ecosystem, population ecology, biodiversity and its conservation, environmental pollution, social issues and the environment, and environmental impact assessment.

BCS ZC223 General Biology (3 units)

Living systems and their properties; major biological compounds; basic physiological processes; introduction to genetics; environment and evolution.

BCS ZC240 General Physics (3 units)

Philosophy of Science; Newton's laws of motion; Work Energy, Impulse and Momentum; Equilibrium; Moment of a force; Rotation; Periodic motion; First law of thermodynamics; Second law of thermodynamics; Electromagnetic waves; Interference and diffraction; Polarization; Relativistic mechanics; Photons, Electrons and Atoms; Quantum Mechanics; Atoms, Molecules and Solids; Nuclear Physics.

BCS ZC111 Basic Electronics (2 units)

Course covers basic passive circuit elements, dependent and independent sources, network theorems, circuit analysis techniques and response of first and second order circuits. Semiconductors - operation of diodes, bipolar junction transistors and field effect transistors. Biasing techniques and transistors. Introduction to operational amplifiers and applications.

BCS ZC113 Online Social Media (2 units)

Social media - evolution, definition, classification, Present social media landscape; Different kinds of online community and social structure; benefits of social media for individuals, business and society; impacts of people and artificial intelligence on social media; Impacts of Social media - personal, professional, social aspects with cases; Concerns (Privacy, authenticity, Informed Consent, Anonymity, Risks), Safety.

BCS ZC114 Video Games - Technology and Social Impacts (2 units)

Video Games- introduction, play and game- deconstructing video games: narrative structures of videogames, critiquing rules in videogames, immersion and materiality- puzzles, drama and socio-technical system of video games- Themes of videogames: violence, addiction, language, sexuality, gender, identity, communities, values and ethics, game genres: e-sports, indie games, serious games-gamification, game design.

BCS ZC313 Introduction to Programming (4 units)

Basic representation of data and how to process data using the representation inside a computer. Techniques for specifying data, operations on data, and problem-solving using C programming language. Systematic techniques and approaches for constructing programs.

BCS ZC228 Introduction to Computing Systems (3 units)

This course teaches a beginner what a computer is and how it can be programmed. The course starts with basic building blocks required to understand the structure of a computer: numbers, bits, structures to implement operations on numbers, and structures to control the execution of commands by a computer. The course progresses to introduce logic gates, and combinational and sequential circuits. It introduces a simple processor design with a small instruction set architecture and basic assembly programming with that processor.

BCS ZC216 Computer Systems and Performance (3 units)

Processor Architecture (Instruction Set Level)- Different types of instructions AL, Memory Access, Branch/Jump; Straight Line Programs vs. Branches: Different Loop Structures, Performance of Loops, Nested Conditionals, Inner Loops – Examples; Cost of Procedure Calls vs. goto statements; Memory Hierarchy: Motivation – Cost vs. Size vs. Access Time. Main Memory and Cache. Write-through vs. Write-back Cache. Access Time Calculation and Hit Ratio; Locality of Reference; Paging (of Main Memory) and Page Faults, Performance Impact; Procedure Calls and Space; Hard Disks and Disk Access (Disk Access Time vs. RAM access Time.); Disk IO and File IO, Buffered vs. Unbuffered IO. Performance: Read vs. Write vs. Read-and-Write; Disk Locality. Sequential Access vs. Random Access in Files - Examples; Flash / SSD Storage: SSD access and access times; Write limitations; Flash File Systems; Performance Impact.

BCS ZC215 Command Line Interfaces and Scripting (3 units)

Files and Directory, internals of a file system, inode structure, block I/O; File commands, pipes and filters, programming with commands; Shell and interpretation of commands, programming with shell scripts, feature usage, interaction with I/Os; Systems calls, usage with C Programming, file system and I/O calls.

BCS ZC311 Data Structures and Algorithms (4 units)

Introduction to Asymptotic Notations, Solving Recurrences (substitution method, iteration method, master theorem), Abstract Data Types, Linear Data Structures (Stacks, Queues, and Linked Lists), Non-Linear Data Structures-Heap, Hash Tables, Binary Search Trees, Balanced Binary Search Trees, Sorting Algorithms (insertion sort, selection sort, merge sort), Graph Data Structures (Adjacency List and Adjacency Matrix), Graph Traversal algorithms, Topological sort, Strongly connected components.

BCS ZC212 Algorithm Design (3 units)

Effective construction and analysis of algorithms. Understanding, application, and implementation of algorithm design techniques like divide-and-conquer, greedy, dynamic programming, and back-tracking. Worst case and average case analysis of algorithms. Basic notions of complexity classes – P, NP, and NP-complete and Reduction.

BCS ZC316 Object Oriented Programming (4 units)

Object orientation concepts, theories and principles; fundamental concepts of the object model: classes, objects, methods and messages, encapsulation and inheritance, interface and implementation, reuse and extension of classes, inheritance and polymorphism; overloading and overriding; static and dynamic binding; multithreaded programming; event handling and exception handling; process of object oriented requirements specification, analysis and design; notations for object-oriented analysis and design; case studies and applications using some object oriented programming languages. Object Oriented Design Patterns: Behavioral, Structural and Creational.

BCS ZC317 Relational Databases (4 units)

Introduction to Database Management Systems; File organization; Data Independence in databases; Data Modeling; Data Definition and Manipulation; Relational Data Model; Relational Algebra & Relational Calculus; Structured Query Languages; Database Design techniques; Functional Dependencies & Normalization; Query processing and optimization; Indexing techniques; Transaction management - concurrency control and recovery; Distributed Databases; Concepts of security and integrity in databases.

BCS ZC234 Programming for Mobile Devices (3 units)

Introduction to mobile computing and emerging mobile application and hardware platforms; Developing and accessing mobile applications; Software lifecycle for mobile application – design and architecture, development – tools, techniques, frameworks, deployment; Human factors and emerging human-computer interfaces (tangible, immersive, attentive, gesture, zero-input); Select application domains such as pervasive health care, m-Health; Mobile web browsing, gaming and social networking; Trends in mobile application development.

BCS ZC238 Web Programming (3 units)

Technologies related to web development and associated technologies that make the web work. Scripting languages like HTML, CSS and JavaScript; Design of dynamic websites; on both client-side and server-side scripting technologies; full-stack web development. Design and development of web applications, web applications that could query database and fetch information over the network; development and testing of web applications.

BCS ZC232 Operating Systems (3 units)

Introduction to operating systems; Various approaches to design of operating systems ; Overview of hardware support for Operating systems; Process/thread management: synchronization and mutual exclusion, inter process communication, CPU scheduling approaches; Memory management: paging, segmentation, virtual memory, page replacement algorithms ; File systems: design and implementation of file systems; Input /Output systems; device controllers and device drivers; Security and protection; Case studies on design and implementation of operating system modules.

BCS ZC236 Software Design Principles (4 units)

Software Development lifecycle; Role of high level and low level design in lifecycle. Object-Oriented Abstraction and Object Oriented Design. Design for Reuse and Design for Change – Refactoring. Design Patterns - History of Patterns in Building Architecture and Relevance to Software Design, Evolution of Software Design Patterns and Impact of Using Design Patterns in Lifecycle. Crosscutting Concerns and Aspects – Aspect Oriented Design. High-level design vs. Low-level design. Basic Architectural Elements and Styles – Layered Architectures and Event-Driven Architectures, MVC architecture in User Interfaces.

BCS ZC214 Building Database applications (3 units)

This course discusses end-to-end application programming involving databases using SQL, PL/SQL, front-end framework, and exposing back-end through APIs. Debugging, Testing, Monitoring, Documenting and Maintaining database applications are discussed. Particular focus will be on Database connectivity; stored procedures; data ingestion/loading; ingestion latency and query performance. This course culminates with a project that involves programming, implementing, and demonstrating a database solution for a business or organization

BCS ZC211 Software Development Practices (3 units)

Review of Use Cases. Design using use cases and data flow; Development Tools and Technologies - IDEs (Eclipse); Compilers and Compiler Options, Libraries and APIs; Code Reviews and walkthroughs; Debugging and Unit Testing, Runtime conditions and Assertions; Runtime Environments (JVM, Android) Code Repositories (GitHub); Versioning; CI/CD.

BCS ZC231 Network Programming and Client Server Programming (3 units)

Overview of computer networks; inter-process communication; network programming; socket interface; client-server computing model: design issues, concurrency in server and clients; external data representation; remote procedure calls; network file systems; distributed systems design.

BCS ZC222 Formal Languages and Applications (3 units)

Regular Expressions, Typical languages expressible using Regular expressions - Regular Languages, Closure Properties; Tokens or Lexemes of Languages (Formal or Natural) – Token descriptions, Scanning Sentences or Programs, Scanner - Design and Implementation. Applications: Specifying patterns, Test cases; Context Free Grammars and Rules, Context Free Languages; Defining the syntax of formal and natural languages; Closure Properties; Limitations of CFGs – features not expressible via CFGs; Context-sensitive Grammars and Languages; Parsing sentences or programs. LL Parsing vs LR Parsing; Look heads - LL(1) parsing, LL(1) parsing engine, and construction of LL(1) parsing table; LL(k) parsing examples and counterexamples, Issues – Ambiguity in CFGs and CFLs. Resolving Ambiguity – Left Factoring and Left Recursion. LR(1) parsing; Application – Natural Language Parsing and POS Tagging;

BCS ZC224 Graphs and Networks (3 units)

Basic concepts of graphs and digraphs, Modelling problems from application domains as graph problem, connectedness, reachability, Euler Tours, Hamiltonian Cycles, planarity, Applications; real world networks: measures (centrality, transitivity, reciprocity), properties and models (random networks, small world model, preferential attachment model), communities in social networks

BCS ZC213 Automata and Computability (3 units)

Finite Automata – DFA, Equivalence of DFAs to regular expressions; PDA – DFA with a stack, equivalence of PDAs to CFLs; Turing Machines – Comparative power of DFAs, PDAs, and TMs. Universal TM; Brief Overview of Chomsky's hierarchy and Church-Turing hypothesis; Non-computable functions; Informal equivalence between TMs and General Purpose computers.

Pre-requisite: BCS ZC222 Formal Languages and Applications

BCS ZC221 Experimental Algorithmics (3 units: 2+1)

Review of time complexity, order of complexity, and Big-O notation; Comparing Theoretical analysis of time and space complexity of elementary algorithms (Sorting, Binary Search trees, Hash tables) with experimental running-time and space measurements. Experiments on key distribution: Measuring heights of BSTs for different datasets and order of arrivals, Uniformity of buckets/bins in hashtables with different hash functions; Size Experiments: Growth of hash table size with datasets and query sequences, Bloom filter size vs. false positive rates, Sketches and cardinality estimation via HyperLogLog. IO Complexity and disk access measurements: Sorting, BSTs vs. B-Trees, Sparse Graphs vs. Dense Graphs. Caching in Data Structures. Case Studies: Use of B-Tree variants in datasets, Bloom filters in Distributed file systems, and HyperLogLog in Redis.

Pre-requisites:

- BCS ZC311 Data Structures and Algorithms
- BCS ZC212 Algorithm Design

BCS ZC227 Introduction to Bioinformatics (3 units)

Course description to be released soon

BCS ZC217 Data Visualization (3 units)

Information overload and issues in decision making. Design of visual encoding schemes to improve comprehension of data and their use in decision making; presentation and visualization of data for effective communication. Elementary graphics programming, charts, graphs, animations, user interactivity, hierarchical layouts, and techniques for visualization of high dimensional data & discovered patterns.

BCS ZC312 Introduction to Data Analytics (4 units)

Introduction to data analytics, applications, Python fundamentals (variables, strings, simple math, conditional logic, for loops, lists, tuples, dictionaries, etc.); Preparing the data - download the data in python, visualizing using matplotlib, seaborn to gain insights, data cleaning, transformation; Regression - applications, fitting models, gradient descent, evaluation, regularized models); Classification - applications, fitting models (decision tree, logistic regression), addressing overfitting, performance, Model selection techniques; Clustering - applications, k-means and hierarchical clustering, quality of clustering; Discussion on advanced topics; Ethical implications of handling data and building models.

Pre-requisites:

- BCS ZC230 Linear Algebra and Optimization
- BCS ZC233 Probability and Statistics
- BCS ZC313 Introduction to Programming

BCS ZC315 Multi-Core Programming and GPGPU Programming (3 units)

Multi-threaded Programming. Review of multicore processors and caching. Shared Memory Programming in the multi-core context. – Use of Locks for Synchronization. Shared Memory data structures - Synchronization issues and models, Lazy synchronization, Design of shared memory data structures. Review of GPGPU architecture and GPGPU programming model. Programming with many-core processors and NUMA. CUDA Programming.

Pre-requisite: BCS ZC216 Computer Systems and Performance

BCS ZC237 TCP/IP and the Internet (3 units)

Review of the OSI model and the Internet model; A top-down approach. Application Layer – HTTP, HTTPS/TLS, DNS, and Multi-media transport; The Internet – Transport, TCP vs. UDP, Routing, and Gateways. IP – Addressing and Routing.

Pre-requisite: BCS ZC231 Network Programming and Client Server Programming

BCS ZC226 Information Security (3 units)

Program security, Web security, Database security, Protection in operating systems, Cloud security fundamentals; Privacy and Anonymity in computing; Legal and ethical issues in security, Secure programming and Trusted systems design; policy, administration and procedures; auditing; physical security; content protection.

BCS ZC225 Human Computer Interaction (3 units)

Principles of human-computer interaction; Evaluation of user interfaces; Usability engineering; Task analysis, user-centred design, and prototyping; Conceptual models and metaphors; Software design rationale; Design of windows, menus, and commands. Voice and natural language I/O; Response time and feedback; Color, icons, and sound; Internationalization and localization; User interface architectures and APIs.

BCS ZC218 Designing Multimodal Interfaces (3 units)

UI Design Principles; GUI Design; Interfaces with multiple modes of interaction – text, graphics, and speech; Identification and authentication: CAPTCHAs; Smart Cards; Design of multi-factor authentication schemes; Identification in Personal Devices – fingerprints, voice-print identification; Design of interfaces for Smart Personal Assistants; Case Study: Siri/Cortana, Alexa/Google Device. Design of chatbots with emphasis on the design of Natural language interaction/conversations.

Pre-requisite: BCS ZC316 Object Oriented Programming

BCS ZC314 Modern Databases (4 units)

Different types of content - Structured vs. Unstructured vs. Semi-structured data, notion of Atomicity (or Consistency) ACID vs. BASE; Strong vs. Weak Consistency model; Serializability; Eventual Consistency; CAP Theorem and Implications; Solution models and Design Examples (some of MongoDB, REDIS, Cassandra, Neo4j); Document stores, Key-Value Stores, Column stores, Graph stores and languages for their query processing, In-memory databases, and cloud databases (E.g. Amazon RDS); Exposure to application development.

Pre-requisite: BCS ZC214 Building Database Applications

BCS ZC229 Introduction to Economics (3 units)

Big ideas in economics; Functioning of the economy; Scarcity and choice; Consumer behaviour; Firm behaviour; Competitive markets; Labour markets; Unemployment; Capital market; Banking and the non-banking financial institution; Central banking; Economic growth; Inflation; Monetary policy; Fiscal Policy.

BCS ZC241T Study Project (5 units)

In this course, students are expected to carry out an organized study and identify a problem which requires a software solution. Students also study the impacts of solving the problem and the existing solutions. At the end of the course, students submit a report with the project proposal, details on background, scope and solution methodology (including requirement specifications).

BCS ZC242T Project (5 units)

The students registered in this course will identify a problem which requires software-based solutions. The students will identify the details of the requirements, design the solution to the problem and implement it using any of the tools/techniques covered as a part of the program. The students will demonstrate their work and submit the detailed project report at the end of the semester.

BCS ZC428T Project (10 units)

In this course, the students work to produce a software solution to a problem whose requirements were studied and documented in the Study Project course. The students will demonstrate their work and submit the detailed project report at the end of the semester.